INT 453: Digital Multimedia

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Lecture 1

Introduction to Digital Multimedia

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books

- Multimedia Technologies & applications for the 21st century.
- Fundamentals of Multimedia.

Agenda

- Introduction
- Concept of Digital Multimedia
- Advantages of Digital Multimedia
- Applications of Digital Multimedia
- Multimedia Today

Introduction

There continues to be a rapid growth in the development and exploitation of multimedia technology. Applications exist in many diverse areas such as:

- commerce
- communications
- education
- entertainment
- art
- scientific research, etc.

Introduction (cont.,)

- The advances in computer technology paved the way to the infrastructure for multimedia applications. With these developments the multimedia concepts have emerged.
- The word multimedia is made up of the two Latin words "multi" which means many and "media" which is the substance through which something is transmitted.

Introduction (cont.,)

In this case, multi is the multiple data types such as

- Text
- Graphics
- Animation
- Sound
- Video

and media is the computer environment used to transmit the information made up of the above multiple data types.

What is Multimedia?

- *Multimedia* involves the combination of two or more media types to effectively create a sequence of events that will communicate an idea usually with both sound and visual support.
- This means that the various sources of media types are integrated into a single system framework.
- Typically, multimedia productions are developed and controlled by computer.

What is Multimedia?

- Multimedia denotes the property of handling a variety of representation media in an integrated manner.
- Multimedia is defined as an interactive computer-mediated presentation that includes at least two of the following elements: text, sound, still graphic images, motion graphics, and animation. (Tannenbaum)

What is Hypermedia?

Hypermedia is another term used related to multimedia.

Hypermedia requires that the user is able to interact with the presentation. The simplest form of hypermedia is hypertext where the user is able to follow textual links.

The most common provider of hypermedia content is the World Wide Web.

What is Multimedia Product ?

- A multimedia product is a presentation, a tool, or an interactive program which is distributed in some kind of a storage medium to be viewed in a suitable computer environment.
- To prepare such a product special tools are also needed. These are referred to as authoring tools.

Evolution of Multimedia

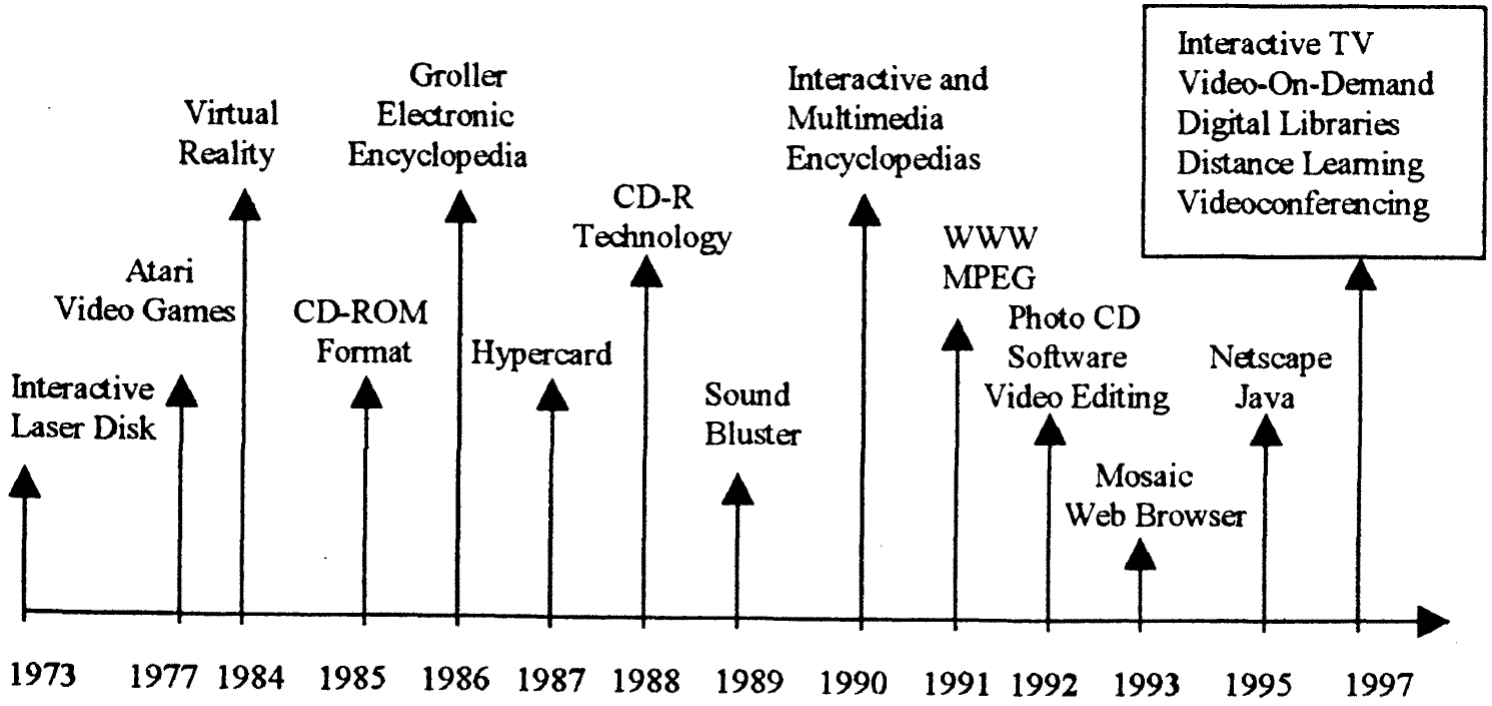


Figure 1. Multimedia roadmap – from 1970s till present time.

Evolution of Multimedia (cont.,)

- Multimedia is not new; the term pre-dates the PC.
- The term has been used for decades to describe slide presentation accompanied by audio tape.
- The combination of slide and narration has been both a popular and successful form of business presentation.

Evolution of Multimedia (cont.,)

In the 1970s the slide show format was introduced to the computer, this technology allowed the computer to control numerous projectors, coordinating them in a manner that produced fast-paced dissolves and effects.

Taped soundtracks would contain cues that triggered the slide projectors to do what it was programmed to do.

Evolution of Multimedia (cont.,)

In the 1980s PCs were designed to "cut" a graphic element and "paste" it into another document. Since then software and hardware developers have been scrambling to integrate various forms of media into the personal computer.

Evolution of Multimedia (cont.,)

- Multi-sensory communication between humans is not a new concept, but is as old as mankind.
- What is new is the “electronic incarnation” of multi-sensory communication that we call multimedia.

Evolution of Multimedia (cont.,)

- The seeds of image projection were sewn as early as 1654 with the “magic lantern”, but further progress in imaging did not occur until 177 years later with the advent of a more ‘advanced” projector called the Phenakistoscope in 1831.
- The next significant development was the cinematograph (movie camera) developed 64 years later in 1895.

Evolution of Multimedia (cont.,)

- In the 55 years between 1895 and 1950, radio, stereo sound and LP records were developed.
- During the 29 years between 1950 and 1979 computer graphics, animations and digital sound were introduced as was ARPANET which was the predecessor of today's internet.

Evolution of Multimedia (cont.,)

- During the 10 years between 1979 and 1989 with the advent of the personal computer and the world wide web, enhancements in multimedia hardware and software were dramatic and the stage was set for even more rapid changes in computer hardware, software and connectivity.
- Between 1989 and 1998 the pace of change has accelerated to the point that hardware and software need to be upgraded every two years just to stay current.

Digital convergence



Digital convergence (cont.,)

- The most important trend in PCs before multimedia was multi-tasking, the ability to run more than one program at a time and you can see why PCs and multimedia are made for each other.
- The enabling force behind multimedia is *digital technology*.
- Multimedia today represents the convergence of digital control and digital media - the PC as the digital control system and the digital media being today's most advanced forms of audio and video storage and transmission.

Digital convergence (cont.,)

- We are in fact experiencing a mass migration from an analog world to a digital world.
- In an analog world communications are accomplished via continuous signals such as those used by telephones for sound and those recorded by VCRs for video.

In a digital world text, images, sound, and video communications are accomplished by sending a stream of digital bits represented by various combinations of zeroes and ones, which (not coincidentally) is the very same technique used to store and manipulate data inside computers.

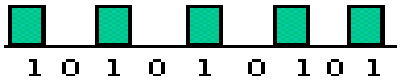
Digital convergence (cont.,)

The illustration below shows a schematic representation of the difference between analog and digital signals and identifies an everyday device that uses these signals.

Digital Source



Digital Signal



Digital Device
CD Player



Analog Source



Analog Signal



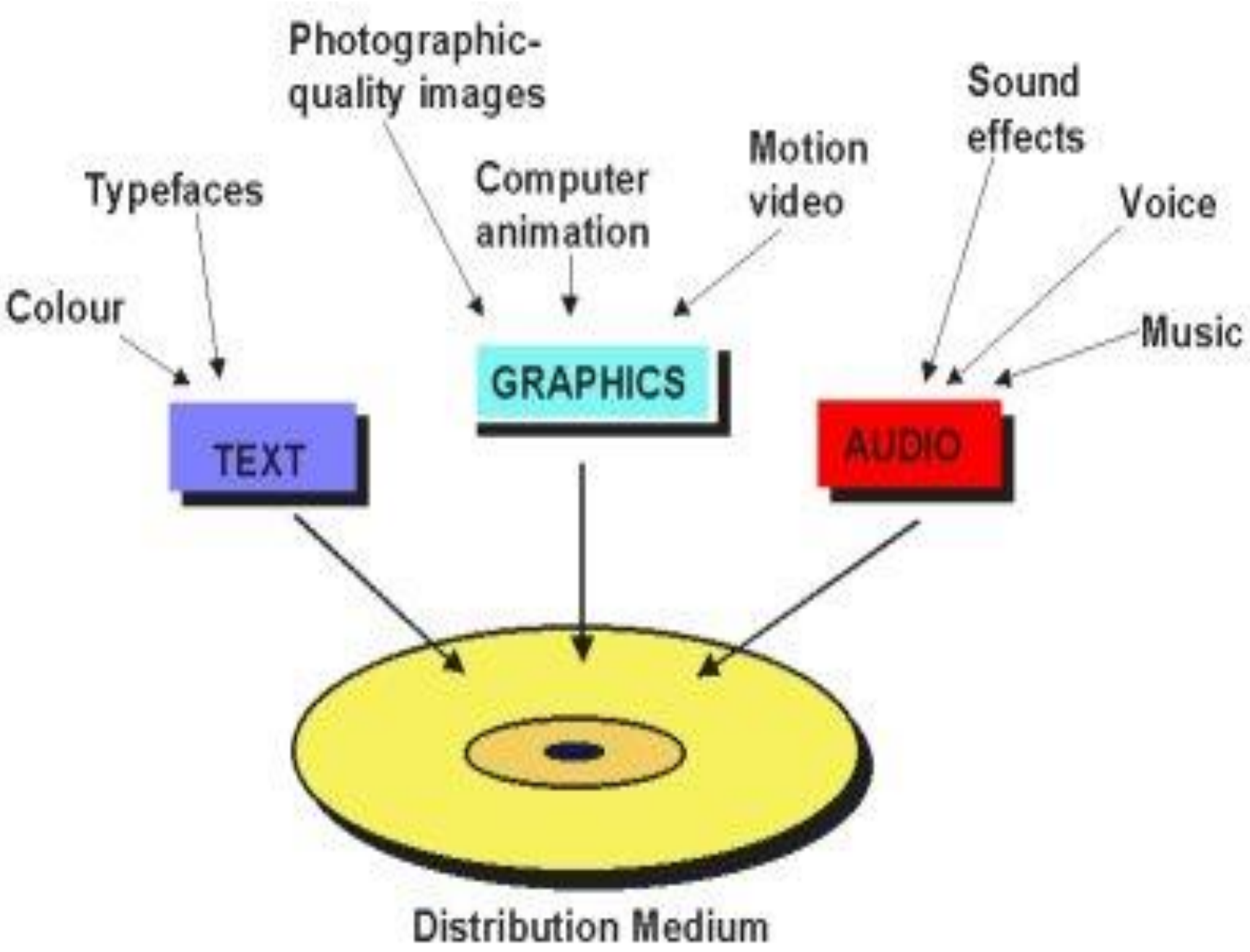
Analog Device

Analog Device



Digital convergence (cont.,)

- Multimedia involves the application of various *communication channels* to a communication exercise. When the various communication channels are used in association with a computer: we call the result computer multimedia.
- The essence of the process of creating multimedia is that a number of types of information (including text, graphics, animation, sound and video) are able to be combined through the use of a computer.


Photographic-quality images
Sound effects
Motion video
Computer animation
Typefaces
Voice
Colour
Music
GRAPHICS
TEXT
AUDIO
Distribution Medium

The advantage of multimedia communications (why?)

- They have the capacity when properly executed,
- To **convey more information more quickly, and more effectively than traditional** communications.
- The effectiveness of communication is an important issue that is at the heart of most human endeavors and is particularly important in the information age where there is frequently an overabundance of information and little time available to sort it all out.

The advantage of multimedia communications (cont.,)

- If a picture is worth a thousand words.
- a moving picture is worth two thousand words.
- a moving pictures combined with text, sound, and direct user involvement via hypertext is worth ten thousand words.
- Multimedia can by definition carry more information, can deliver it more quickly and can deliver it in a form that builds on what the user already knows and therefore requires less user interpretation and less user time to understand more of the message being sent.

The advantage of multimedia communications (cont.,)

- In its simplest terms every communication requires a sender, a message to be sent, and a receiver.
- The addition of a digital component such as sound, clipart, photographs, animated characters, video clips or interactive navigational mechanisms results in a multimedia communication.
- In this day and age senders and receivers use machines to communicate.

The advantage of multimedia communications (cont.,)

Machines used in electronic communications include telephones, modems, computers, data collection devices such as electronic cash registers, electronic time clocks, grocery store optical scanners, televisions, VCRs, camcorders, digital cameras, stereo receivers and video-teleconferencing equipment.

Regardless of the nature of the communicators, the effectiveness of the communication remains a function of the *quantity of information communicated, speed of the communication*, and the *degree of comprehension* of the communication's meaning.

Multimedia communications

include television advertisements, computer-based business, and classroom presentations using programs such as

- Microsoft PowerPoint©,
- World Wide Web (WWW)
- home pages, and
- a wide range of products including electronic games, edutainment (programs that are both entertaining and educational such as learn to read and learn to count),
- corporate training programs,
- flight simulators and
- sales presentations that are distributed via media such as
  - diskettes and
  - CD ROMs.

Hypertext and Hypermedia

HYPERTEXT is usually defined as NON-LINEAR ACCESS TO TEXT using links embedded in the text This way going through material that is not of interest is avoided.

HYPERMEDIA is the extension of this non-linearity to other media types.

Multimedia Assets (How?)

Effective multimedia applications depend on the most effective use of various materials – referred to as *assets* or resources.

1- Text

Text constitutes the main part of a multimedia package. It is used to provide most of the information intended to be conveyed and it is even stated that other multimedia data types are used to enhance text.

There are 3 major advantages generally associated with screen-based text compared with paper-based text:

- the ability to spontaneously update the screen,
- the reactive capability, and
- the ability to incorporate special effects.

Text (cont.,)

However, there are also disadvantages:

- text is much harder to read from a screen than it is from paper.
- people tend to print copies of information they are accessing from the computer. This may be due to habits or the effect the current displays have on the eyes.

Using media other than text may require extra resources compared to text but definitely makes it much easier to follow the material covered.

2- Audio

Sound is another data type used in multimedia applications. It can add a particular dimension of reality to multimedia systems. Sound requires more space than text but is better when compared to video clips.

People use it to express an idea in words over the computer or it can also be used to introduce effects into a presentation. Either way it ENHANCES THE QUALITY of presentation and INCREASES THE EFFICIENCY of information transfer.

3- Images

- Images improve the overall look of a presentation, and they are useful to express information text alone cannot convey.
- Using IMAGES and GRAPHICS can be very useful but it has to be noted that they also introduce extra load to the system both as storage and also as network traffic.

3- Images (cont.,)

There are six different image formats:

- GIF files
- JPEG files
- Animated GIF files
- MPEG files
- Shockwave files
- NxView files

3- Images (cont.,)

- Images come in different forms and resolutions i.e. the number of picture elements used to represent them (pixels) and they are usually compressed using different techniques such as JPEG.
- JPEG is a standardized image compression mechanism.
- It stands for Joint Photographic Experts Group, the original name of the committee that wrote the standard.

3- Images (cont.,)

When you are sending digital photographs as email attachments or trying to pack as many images on a floppy disk as possible, the size of each image is important. What you want to do is reduce the file size to as few bytes as possible without hurting the image quality.

Most websites use the JPEG format for shrinking images. JPEG is a popular format for two reasons:

- It can make image files smaller. (It lets you adjust the amount of compression.)
- It stores 24-bit-per-pixel color data instead of 8-bit-per-pixel data. (It has good compression characteristics on photographic data.)

3- Images (cont.,)

- There are various image formats. However, the most widely used formats are gif, and jpg.
- Both of these formats has static (as opposed to animated) bitmap images.
- In a bitmap image, the image file has to define the exact color of every pixel in the image. For example, imagine a typical bitmap on the web that is 400 by 400 pixels.
- To define this image, you would need 24 bits per pixel for 160,000 pixels, or 480,000 bytes. That would be a huge image file, so both the GIF and JPG formats compress the image in different ways.

3- Images (cont.,)

In a GIF image, the number of colors is reduced to 256 and then "runs" of same-color pixels are encoded in a color+number of Pixels format.

For example, if there are 100 pixels on a line with the color 41, the image file stores the color (41) and the length of the run (100). This makes a GIF file great for storing drawings that have lots of same-color pixels.

4- Video

A video clip or an animation sequence can express an idea in a much better way than text or images. Video clips used in multimedia systems are more flexible in the sense that they are EASIER TO EDIT and also EASIER TO ACCESS.

4- Video

DIGITAL VIDEO can be created from VCRs (a videocassette recorder), using cameras or directly recorded from broadcasts. Either way they are captured, stored and edited before they are used in multimedia presentations.

Portable video recorders have made it relatively easy to capture real-time video images.

Video can be incorporated into multimedia applications using two different processes.

- The first involves using a video source connected to the computer via a controller card. This technology has been referred to as **interactive video**.

4- Video (cont.,)

- A more integrated process converts video from analogue into digital format that can be manipulated by the desktop computer. This approach (**called *digital video***) allows all video operations — including editing and special effects — to be carried out on the computer.

Video is usually played at 25 or 30 frames per second and will have to be COMPRESSED in order to reduce it to acceptable sizes. Different compression techniques are used and MPEG is one of these methods.

4- Video (cont.,)

ANIMATION is similar to video, in that it uses the display of moving pictures to convey information. The pictures are constructed artificially, however, and they can be very useful to explain abstract concepts.

They also require huge storage space and packages used to produce them are usually difficult to learn.

4- Video (cont.,)

- Video files are perhaps the most exciting and attractive features of multimedia systems. The potential to include complex graphics and video images in multimedia programs is a great strength of multimedia technology.
- Words are almost useless for conveying information from those who understand to those who don't, although words are an excellent means of exchange between people who understand equally well.
- A word has a unique meaning; pictures, however, transcend language codes and people extract their own meanings from pictures.

4- Video (cont.,)

- Multimedia software allows developers to organize and direct the combination of text, graphics, animation, audio and video images to produce multimedia programs.
- The ability to apply the best communication tools for each component of a communications problem, and to do so in a cohesive way, is really the essential strength of any multimedia presentation.
- Here are some examples of various types of video files. In order to view these you may need some PLUG-IN's.
- avi file example
  mpg file example

Multimedia Applications

Multimedia can be used

- to inform,
- to sell,
- to teach,
- to persuade and/or
- to entertain.

Multimedia Applications (cont.,)

Beyond these general purposes, digital multimedia enables the application of technology to a wide range of specific things for business and pleasure. These include

- electronic brochures,
- marketable games,
- orientation and training programs,
- distance learning,
- information kiosks,
- electronic resumes,
- membership recruiting,
- electronic business presentations,
- web site creation, and
- “small-screen” productions including advertisements and movies.

Electronic brochures

can be used for advertising and selling products, ideas and events; for presenting policies, procedures and techniques; for teaching everything from automotive repair to computer software and music. These brochures can be distributed over the web and/or can be burned (copied) onto a compact disk and distributed personally as in an automobile showroom or in the mail.

Marketable games

created with multimedia can provide everything from entertainment to edutainment and can be created for the young, the not so young, and everyone in between. Games can be created to pit a single or multiple players against each other, a computer, or some combination of these.

Games can be classified into six primary categories which include

- 3D simulations,
- card games,
- board games,
- survival games,
- mind games, and
- educational games.

Distance learning

- Distance learning can support continuity of learning even when the learner and teacher are physically removed from each other.
- Multimedia can be used to create, package and distribute course content.
- In this environment the design and execution of multimedia elements has direct impact on the quality of programs delivered.

Distance learning (cont.,)

- But one of the greatest challenges to educators is the diversity of students, especially in the different ways they learn. Some students learn better through association, others by experimentation, some are more visually oriented, others are more auditory.

- Multimedia has the ability to accommodate different learning styles and can present material in a non linear manner. It is motivating, it can be highly interactive, it can provide feedback and evaluate skills. And it can make learning fun.

Orientation & Training

- titles can be used for everything from introducing workers to their new employer's facilities and people, to training highly experienced jet engine mechanics on how to fix a newly designed engine component.
- The combination of video, sound, animation, and hypertext can expedite the learning process and leverage the time of instructors.
- "Virtual" or cyberspace learning can also have significant cost and safety advantages. Jet pilots, for example, can "practice" flying in a multimedia training simulation without risking the loss of a multimillion dollar aircraft and many human lives.

Information Kiosks

are becoming a mainstream vehicle for providing directions and general information services that have traditionally been provided by signs or by people.

Multimedia Kiosks can be found

- in the entryways of shopping malls,
- in interstate highway rest areas,
- in museums,
- in airports,
- in hotels,
- in amusement parks,
- in casinos and
- virtually everywhere that large numbers of people seek directions or information.

Information Kiosks

- Interactive multimedia kiosks can be used to give highway directions, to explain how something is done such as how wine is made, to show the location of particular geographical points of interest, or to provide status updates on dynamic things such as stock market activities, late breaking news, road conditions, and the weather.
- For better or for worse, multimedia kiosks are becoming effective substitutes for human guides. They don't take breaks and they can work 24 hours per day. In many cases where the information being provided is particularly well suited to communication via multimedia, the communication is faster and more effective, not to mention considerably less expensive.

Electronic resumes

- New phenomena that enable both employers and prospective workers using the Web to quickly search wider geographical areas for the perfect match.
- For the student or individual seeking a position, an electronic resume uses multimedia to showcase speaking, writing, technical and specialized skills and interests.
- It can provide extensive distribution to potential employers that far exceeds the number of resumes that would normally be photocopied and mailed by an individual or by a placement agency.

Membership recruiting

- using digital multimedia and distributed via CD ROM or the Web is used by governmental, religious, political, environmental, business and private organizations to garner members, votes and ideas.
- Multimedia can be used to describe their specific programs and objectives and to provide membership documents. A classic example is that of public broadcasting stations' solicitation of donations.

Electronic business presentations

- Used for communicating, selling, analyzing, and educating. They have evolved from traditional speech delivery to speeches supported by overhead projectors and are now speeches supported by notebook computers and portable color projectors.
- The advent of powerful multimedia-capable notebook computers loaded with presentation graphics software enables presenters to become self-contained production studios.
- Their presentations are custom tailored to the particular needs of their audiences and are presented quickly, efficiently and effectively.

Multimedia Software

The following software programs progress from basic presentation to complete authoring capabilities:

- Microsoft PowerPoint
- Aldus Persuasion
- Adobe Premiere
- Macromedia Director
- Macromedia Authorware
- Asymmetrix ToolBook II Assistant

Web site creation

- has become a prime activity for the application of multimedia. This is due in large part to the nature of Web broadcasting or Webcasting which relies on hypertext to enable users to navigate.
- Home pages can be customized to reflect the message and personality of an individual or organization.
- Software to create Web-pages has evolved from hard to use plain text editors based on HyperText Markup Language (HTML) codes to graphical based editors that can use objects and formatted text. HoTMetal Pro, Microsoft Front Page and Netscape Composer are examples of Web-Page Editors.

“Desktop” productions

- Emerging applications for multimedia distributed via the Web and CD ROM. These are advertisements, movies and electronic short stories that directly combine computer, entertainment and communications resources to replicate entertainment and edutainment facilities that were historically the exclusive domain of movie and television studios.
- This application of multimedia is in its infancy, yet it has the potential to “democratize” an industry that is very capital intensive and therefore to open creative and financial opportunities to anyone with a creative bent who has access to a computer, multimedia software and the Web.

Edutainment

Drawing the line between education and entertainment in multimedia can be almost impossible, hence the term 'edutainment'. Multimedia *can* make learning entertaining.

But multimedia also has a purely entertainment side. Anything that's possible in sound and images is possible on a multimedia CD.

AIATSIS

AIATSIS is an encyclopedia of the Australian Aborigine containing over 2,000 entries - 1,000 photos, 230 sound clips and 50 videos. It covers subjects ranging from art to health, from technology to law.

World Architecture is a collection of 5 CD disks containing thousands of images of architecture throughout the world.

Virtual Reality simulations

and the application of this technology to education, training and entertainment are an evolving specialty area within multimedia.

Virtual Reality has been defined as consensual hallucination and can be viewed as three dimensional multimedia that can be delivered via special devices such as electronically equipped chairs, helmets, glasses, and gloves.

Virtual reality stimulates human perceptions of touch, motion and depth.

Ten Reasons to Use Multimedia

PC WORLD MAGAZINE (OCTOBER 1993) OUTLINED "TEN REASONS TO USE MULTIMEDIA IN EDUCATION." WHILE IN MANY CASES UNPROVEN AND OVERSTATED, THEY ALSO UNDERSCORE THE POTENTIAL OF MULTIMEDIA IN EDUCATION.

fast -- learning speed accelerates.

cheap -- the program never asks for a raise; the more you use it, the less it costs per use.

consistent -- no mood swings, yawns, or lapses.

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private -- ask what you want; no one will laugh, no one will scold.

safe -- experience nuclear meltdowns without fallout; experience drunk driving accidents or electrocution without blackouts or death.

personal -- it never tires of praising and motivating through positive feedback, any time, day or night.

a strong foundation -- on which to build mastery.

a tool to make remembering longer, easier -- many parts of the brain are stimulated.

more information faster -- on things a school couldn't afford to teach: like space-shuttle repair, brain surgery, black hole sailing.

fun -- like a game: yes, like Nintendo, which, with a joystick and a screen, has already captured the brains and fingers of an entire generation.

Delivering Multimedia

CD ROM

The growth of multimedia, as mentioned, is often expressed in terms of the growth of compact disk titles. CDs are a popular medium because they can hold substantial amounts of data, including sound and video. They are relatively inexpensive; easy to mass produce, distribute, and transport; and they take up little retail shelf space. In addition, it is now common for computers to be sold with CD ROM drives, sound cards and speakers.

KIOSK

A *kiosk* is a stand-alone or networked computer system that allows the user to access information, perform transactions, and even play games. Examples include

- University information kiosks that students use to learn about academic programs, print out schedules and transcripts and access campus maps;
- Retail store kiosks that allow customers to locate merchandise, print out coupons and purchase products;
- Gallery kiosks that allow the user to locate specific works of art, view parts of a collection not on display and obtain detailed information about artists.

KIOSK

Kiosks are useful in disseminating information especially in high traffic areas, providing value added services to customers (convenience), and reducing personal costs. Kiosks can be expensive because of the investment in hardware and the need to continually update their contents.

ONLINE

- Obviously the fastest growing area for multimedia delivery is online - which includes telecommunications and the Internet.
- Telecommunications involving phone lines, satellite and cable transmission is used by educational institutions to deliver multimedia courseware and by companies for teleconferencing and training.
- The use of the Internet is expanding in all areas. Companies are now commonly using the WWW to allow customers to purchase products, access product information and subscribe to real time multimedia events such as rock concerts.

Multimedia Constraints

Multimedia constraints are situations that bound what can physically be done with the creation and distribution of multimedia. These include the following:

1. Effective multimedia is not as easy to create as some would have us believe. Effective multimedia accomplishes its stated purpose and is delivered succinctly. Creating such multimedia takes considerable time and skills and is seldom the result of a "shot in the dark" approach;
2. Multimedia requires high powered software and hardware engines and at the same time have kept costs down to very reasonable levels.

Multimedia Constraints (cont.,)

3. The very large physical size of digital multimedia elements and titles causes significant data capture, storage, and playback problems. The capture and storage in digital form of a single 10 second video clip may require up to 100 Megabytes of storage space on your computer. By extension one minute may require 600 Megabytes. Such requirements become massive when considering a multimedia title such as a full length movie.
4. Multimedia distribution channels and standards are in a state of instability. Traditionally movies were delivered in theaters. More recently they have been released in VHS tape format for home use on VCRs. Now, Digital Video Disks that can store a whole movie with superior visual and sound quality.

How to identify and protect against some of the risks involved in using Multimedia

Technology

- You need to practice **basic safety techniques - back up** frequently to incrementally named files, isolate faulty components, don’t perform lengthy tasks until simple diagnostic checks are exhausted.
- You need to **guard against wasting your time** in many other ways.
- You **need to keep all media components as small as** they can be possible, and to identify tasks which would stretch a system to failure or partial malfunction.

Technical Demands

- A multimedia system can store audio and video information and use it later as in the case of training or transmitting it.
- Live in real-time. Transmitting live multimedia information imposes many constraints on multimedia systems.
- **The NETWORKS SHOULD BE HIGH SPEED** to cope with the large amounts of data to be transmitted synchronously over them.
- The **PROCESSORS must be powerful** enough to execute the software fast enough.

Technical Demands

- The BUS must have enough bandwidth. Communication techniques and requirements are also different to those commonly employed today.
- **Operating system issues have to be handled**.
- Data types are diverse and different techniques for handling these are required eg. video compression.
- Special tools for producing multimedia applications are required.
- There must be sufficient and fast storage capacity. Information retrieval techniques suitable for the new types of data need to be developed.

Inappropriate use of multimedia

As mentioned earlier, despite its obvious advantages, it is not always appropriate to use multimedia.

Considering the future

Multimedia is still a new, rapidly advancing, field of technology and communications. It will generate many opportunities both for taking existing on-line services forward and creating new ones.

At present some of the 'nice to have' features of the interface can be more hindrance than help. An obvious example is video: **its quality, the size** of the window and machine speed are all current issues.

Given the current state of technology, **websites and multimedia applications will need regular updates**.

Considering the future

You want to use multimedia technology?

You want to play with the latest and greatest toys, and you want to change the world while you're at it. You can do it. It's all there for you.

No-one else has more opportunity than you to do something that hasn't been done before.

Within a few weeks you will be at the "bleeding edge". You will need to understand the basic technology, know about the standards that affect the hardware, and avoid getting sucked into being a computer technician.

Today's lecture is both an introduction to the module itself and to the theory behind it.

Review Questions

1. What is the difference between analog and digital multimedia?
2. What is the difference between multimedia, hypertext, and hypermedia?
3. Explain 5 applications of Digital Multimedia?
4. What is Edutainment, and how you can use it?
5. Explain the basic Multimedia Constraints?
6. How to protect against some of the risks involved in using Multimedia?

Questions

Thank you